

TASK-10

7-7-2026

TOPIC : INSTALL THE NGINX AND APACHE WEB SERVER

“INSTALL THE NGINX WEB SERVER”

Step 1: Connect to your EC2 instance

```
ssh -i "kittu.pem" ubuntu@ec2-3-81-129-45.compute-1.amazonaws.com
```

Step 2: Update the package list

Command: **sudo apt update**

```
ubuntu@ip-172-31-33-150:~$ sudo apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease [136 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 Packages [16.3 MB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
```

Step 3: Install Nginx

Command: **sudo apt install nginx -y**

```
ubuntu@ip-172-31-33-150:~$ sudo apt install nginx -y
Installing:
  nginx

Installing dependencies:
  nginx-common

Suggested packages:
  fcgiwrap nginx-doc ssl-cert

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 0
  Download size: 664 kB
  Space needed: 1876 kB / 4258 MB available

Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 nginx-common all 1.28.3-2ubuntu1.6 [37.3 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 nginx amd64 1.28.3-2ubuntu1.6 [627 kB]
Fetched 664 kB in 0s (19.3 MB/s)
Preconfiguring packages ...
```

Step 4: Check the Nginx service status

Command : **sudo systemctl status nginx**

```
ubuntu@ip-172-31-33-150:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-07-07 03:09:57 UTC; 22s ago
     Invocation: 2de263ee470a43alb4271d368b3d8cda
       Docs: man:nginx(8)
    Process: 17844 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
    Process: 17846 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 17880 (nginx)
      Tasks: 3 (limit: 627)
     Memory: 3M (peak: 7.1M)
        CPU: 45ms
    CGroup: /system.slice/nginx.service
            └─17880 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
               └─17882 "nginx: worker process"
                  └─17883 "nginx: worker process"

Jul 07 03:09:57 ip-172-31-33-150 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Jul 07 03:09:57 ip-172-31-33-150 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
```

To Edit the Inbound Rules

Steps:

1. Click Instances.
2. Select your EC2 instance.
3. Scroll down to the Security tab.
4. Click Edit inbound rules.

Inbound rules	Outbound rules	Sharing	VPC associations	Related resources - new	Tags
Inbound rules (1)					
Search					
☐	Name	Security group rule ID	IP version	Type	Protocol
☐	-	sgr-0b543cc1a1f8ae1b0	IPv4	SSH	TCP

5. Enter the following values:
6. Click Save rules.

Inbound rules Info

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info
sgr-0b543cc1a1f8ae1b0	SSH	TCP	22	Cus...	
-	HTTP	TCP	80	An...	

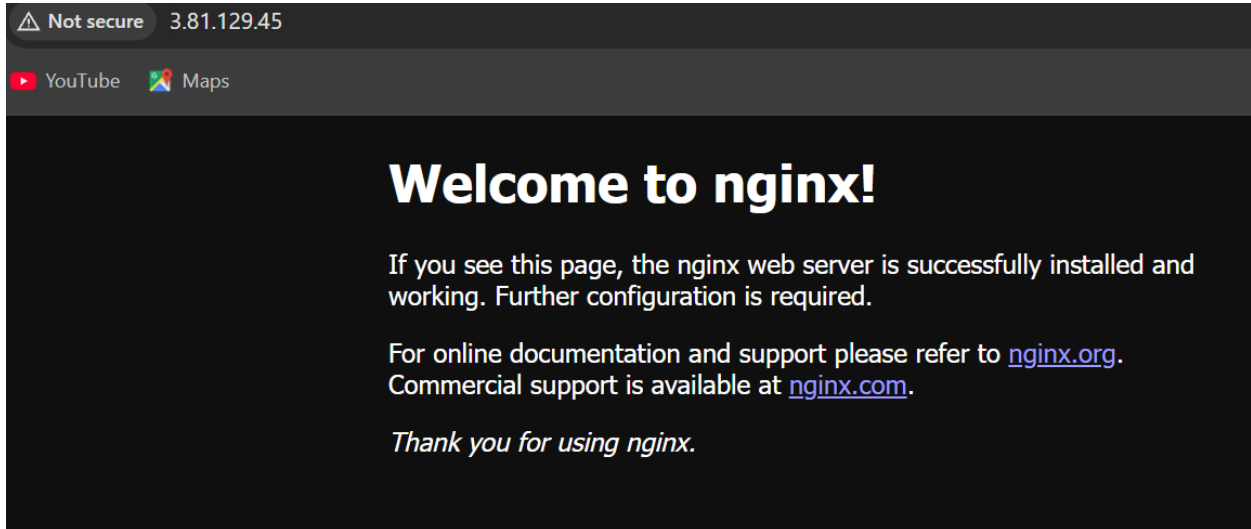
Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

✓ Inbound security group rules successfully modified on security group (sg-0ffd7e90bed721390 | [launch-wizard-7](#))
▶ Details

To see the website:

Copy the public ip address: **3.81.129.45**

Open google paste the ip address



Enter the commands:

cd /

cd var

cd www

cd html

ls

```
ubuntu@ip-172-31-33-150:~$ cd /
ubuntu@ip-172-31-33-150:/$ cd var
ubuntu@ip-172-31-33-150:/var$ cd www
ubuntu@ip-172-31-33-150:/var/www$ cd html
ubuntu@ip-172-31-33-150:/var/www/html$ ls
index.nginx-debian.html
ubuntu@ip-172-31-33-150:/var/www/html$ |
```

Create the file:

Command : **sudo touch index.html**

ls

```
ubuntu@ip-172-31-33-150:/var/www/html$ sudo touch index.html
ubuntu@ip-172-31-33-150:/var/www/html$ ls
index.html  index.nginx-debian.html
```

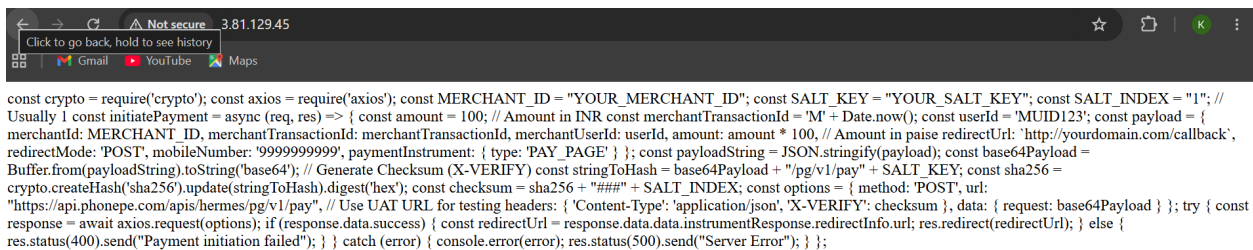
To write the code on that:

Command : **sudo vi file name** after the command click the **"I"**

1. To insert the code
2. Take code from the google what type code you need copy the code paste the code in the terminal
3. After that click "ESC" and enter the command "**:wq!**" to save the code

```
console.error(error);
res.status(500).send("Server Error");
}
;
wq!
```

Open the google and refresh the page



The screenshot shows a web browser window with a "Not secure" warning and a status bar indicating the address is 3.81.129.45. The browser's address bar shows a "Click to go back, hold to see history" message. Below the address bar, there are search engines (Google, YouTube, Maps) and a list of bookmarks. The main content area displays a JavaScript error: "const crypto = require('crypto'); const axios = require('axios'); const MERCHANT_ID = 'YOUR_MERCHANT_ID'; const SALT_KEY = 'YOUR_SALT_KEY'; const SALT_INDEX = '1'; // Usually 1 const initiatePayment = async (req, res) => { const amount = 100; // Amount in INR const merchantTransactionId = 'M' + Date.now(); const userId = 'MUID123'; const payload = { merchantId: MERCHANT_ID, merchantTransactionId: merchantTransactionId, merchantUserId: userId, amount: amount * 100, // Amount in paise redirectUrl: 'http://yourdomain.com/callback', redirectMode: 'POST', mobileNumber: '9999999999', paymentInstrument: { type: 'PAY_PAGE' } }; const payloadString = JSON.stringify(payload); const base64Payload = Buffer.from(payloadString).toString('base64'); // Generate Checksum (X-VERIFY) const stringToHash = base64Payload + '/pg/v1/pay' + SALT_KEY; const sha256 = crypto.createHash('sha256').update(stringToHash).digest('hex'); const checksum = sha256 + '###' + SALT_INDEX; const options = { method: 'POST', url: 'https://api.phonepe.com/apis/hermes/pg/v1/pay', // Use UAT URL for testing headers: { 'Content-Type': 'application/json', 'X-VERIFY': checksum }, data: { request: base64Payload } }; try { const response = await axios.request(options); if (response.data.success) { const redirectUrl = response.data.data.instrumentResponse.redirectInfo.url; res.redirect(redirectUrl); } else { res.status(400).send("Payment initiation failed"); } } catch (error) { console.error(error); res.status(500).send("Server Error"); } };

To remove the nginx server

Commands: **sudo systemctl daemon-reload**
sudo systemctl disable nginx
sudo apt remove nginx -y

```
ubuntu@ip-172-31-33-150:/var/www/html$ sudo systemctl daemon-reload
ubuntu@ip-172-31-33-150:/var/www/html$ sudo systemctl disable nginx
Synchronizing state of nginx.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install disable nginx
Removed '/etc/systemd/system/multi-user.target.wants/nginx.service'.
ubuntu@ip-172-31-33-150:/var/www/html$ sudo apt remove nginx -y
The following package was automatically installed and is no longer required:
  nginx-common
Use 'sudo apt autoremove' to remove it.

REMOVING:
nginx
```

“INSTALL THE APACHE WEB SERVER”

After the removing the nginx again the page was not update means try the commands

Install Apache

sudo apt install apache2
sudo cp /usr/share/apache2/default-site/index.html /var/www/html/index.html
Sudo systemctl restart apache2

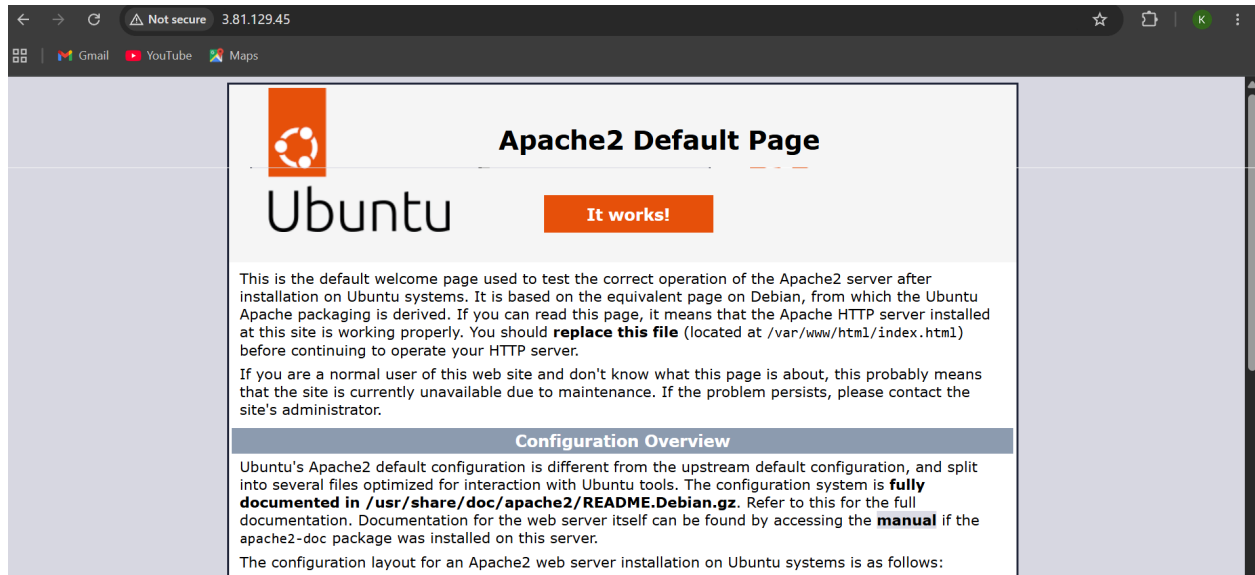
```
ubuntu@ip-172-31-33-150:~$ sudo apt install --reinstall apache2 -y
The following package was automatically installed and is no longer required:
  nginx-common
Use 'sudo apt autoremove' to remove it.
```

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-33-150:~$ sudo cp /usr/share/apache2/default-site/index.html /var/www/html/index.html
ubuntu@ip-172-31-33-150:~$ sudo systemctl restart apache2
```

To see the website:

Copy the public ip address: 3.81.129.45

Open google paste the ip address



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ls

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ubuntu@ip-172-31-33-150:/var/www/html$ ls
index.nginx-debian.html
ubuntu@ip-172-31-33-150:/var/www/html$ |
```

Create the file:

Command : **sudo touch index.html**

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ubuntu@ip-172-31-33-150:/var/www/html$ ls
index.html  index.nginx-debian.html
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```
console.error(error);
res.status(500).send("Server Error");
}
;
:wq!
```

Open the google and refresh the page



Phone Manufacturer

Apple

▼

Show Manufacturer

Manufacturer: Apple

